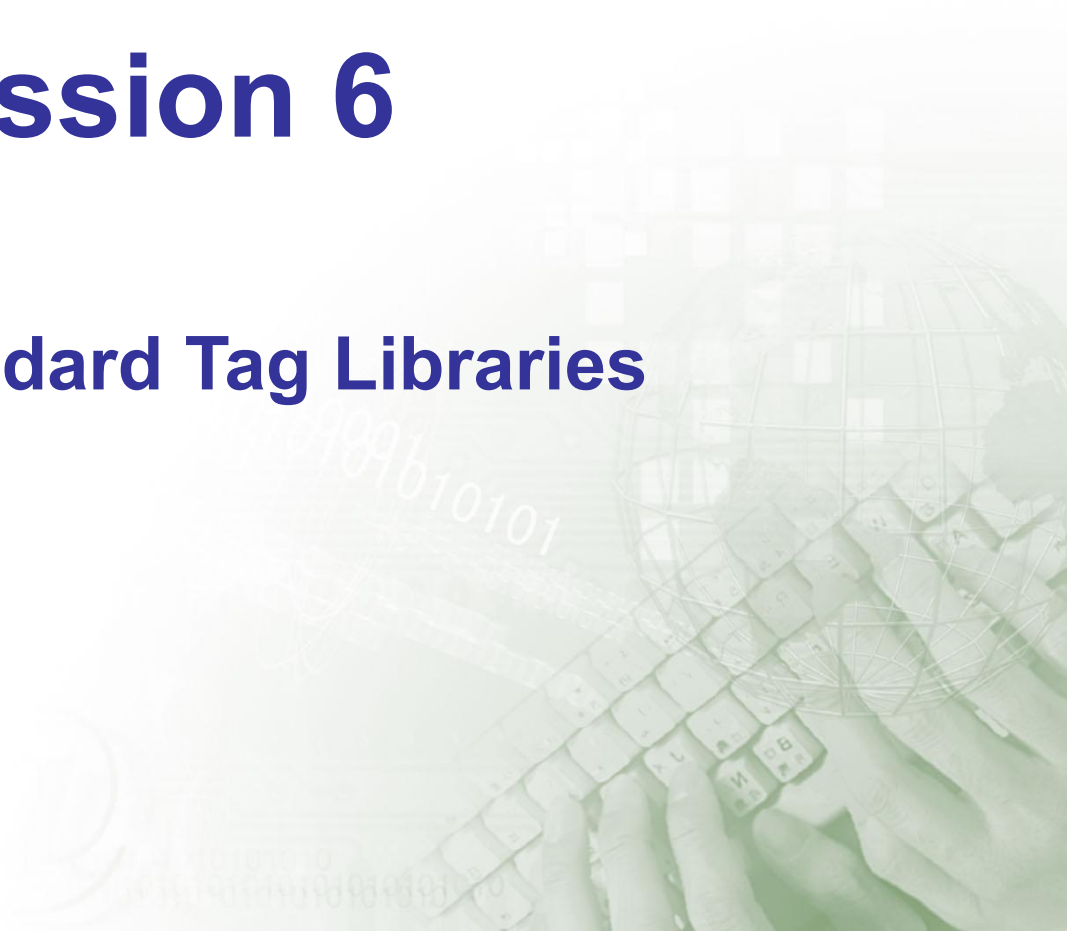


Session 6

JSP Standard Tag Libraries



Review

- ❑ Custom tags embed small bits of Java code in a JSP page and follow standard HTML tag format.
- ❑ Custom tags can be implemented by creating either a Java class or a tag file.
- ❑ Custom tags implemented using a Java class file:
 - ❑ Require a tag handler and a Tag Library Descriptor file.
 - ❑ Have to extend or implement the `javax.servlet.jsp.tagext` package.
- ❑ Custom tags implemented using tag file:
 - ❑ Do not require a Tag Library Descriptor
 - ❑ Can be used with JSP engine 2.0 and above
- ❑ Tag library helps group custom tags with similar functionality.

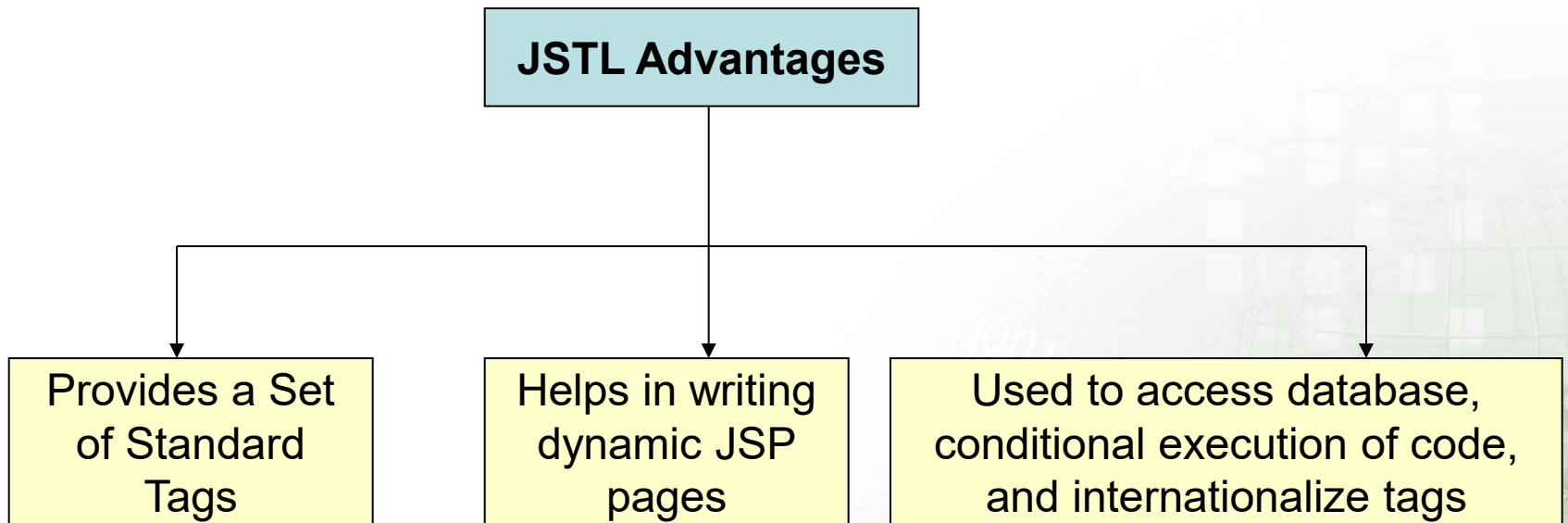
Objectives

- ❑ *Explain Core tag Library*
- ❑ *Explain I18n and Formatting Tag*
- ❑ *Explain SQL tag Library*

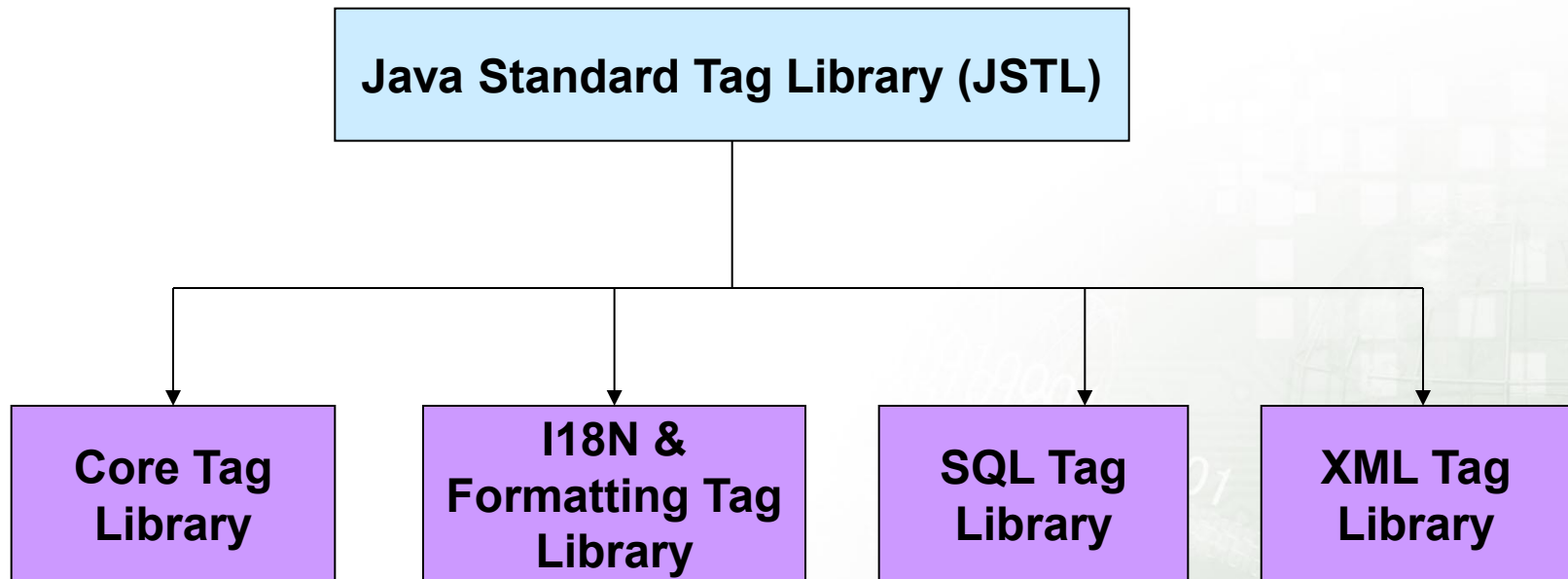
- ❑ The JSP Standard Tag Library (JSTL) represents a set of tags to simplify the JSP development.
- ❑ **The Main advantages:**
- ❑ Fast Development JSTL provides many tags that simplify the JSP.
- ❑ Code Reusability We can use the JSTL tags on various pages.
- ❑ No need to use scriptlet tag. It avoids the use of scriptlet tag.

Introduction

- ❑ Java Server Pages Standard Tag Library (JSTL) contains tags that can be used in a JSP page.



- ❑ JSTL provides four main tag libraries:

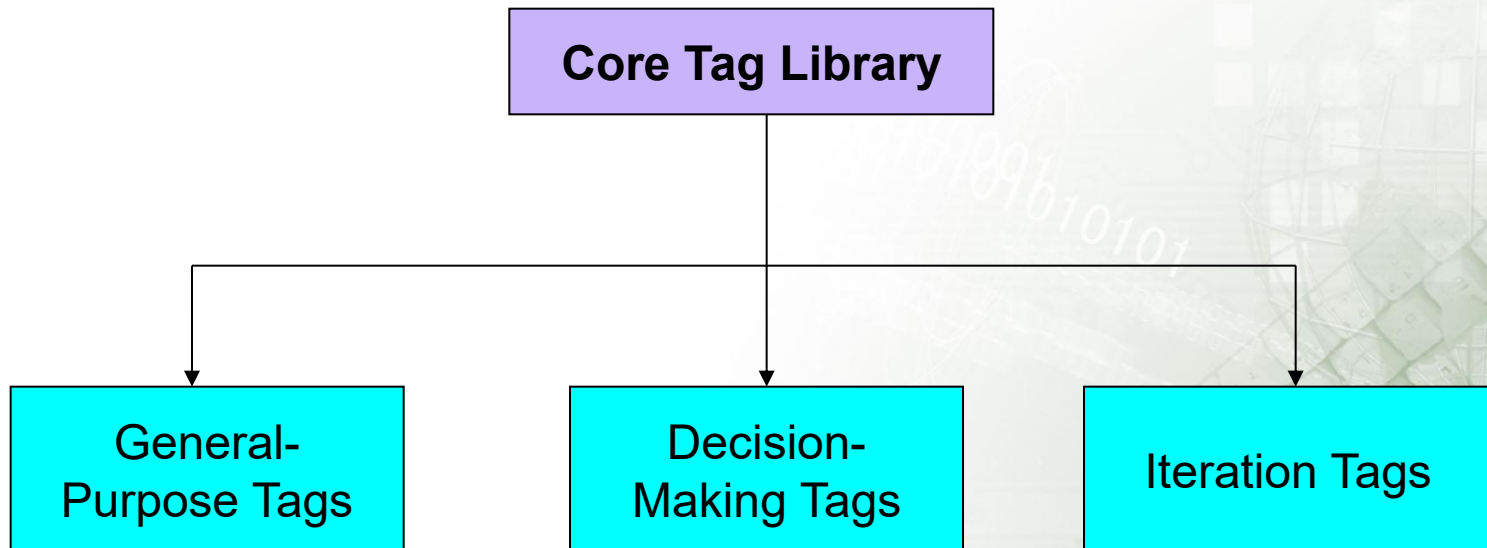


Core Tag Library

Syntax:

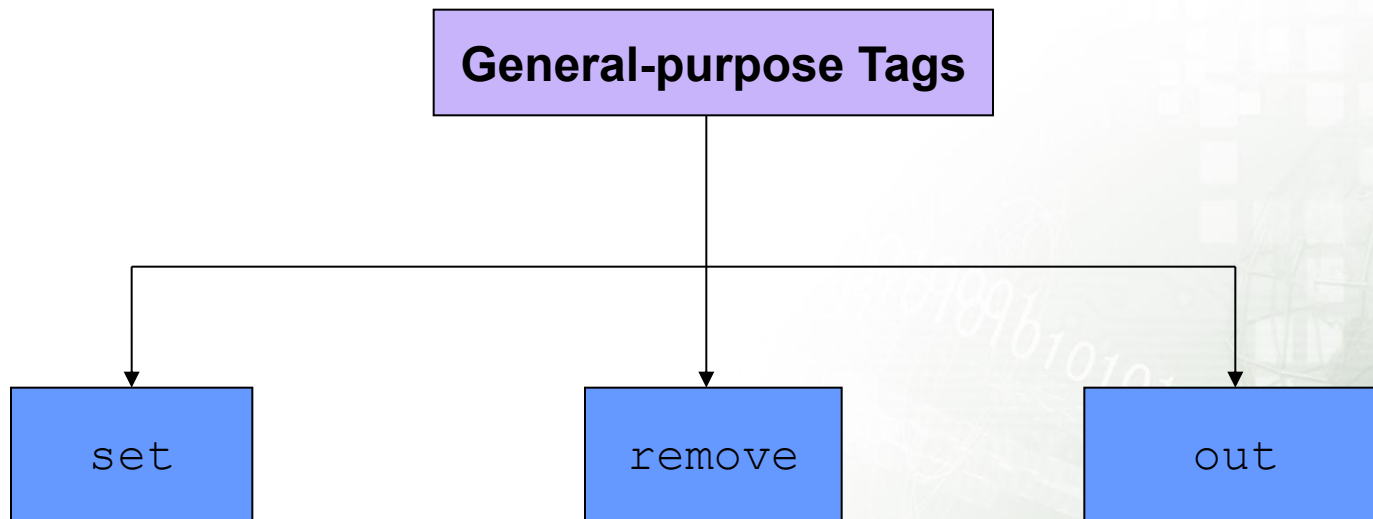
```
<%@ taglib uri =  
    "http://java.sun.com/jstl/core_rt"  
    prefix = "c" %>
```

- ❑ Core Tag Library has the following tags:



General-purpose tags 3-1

- ❑ General-purpose tag library is used to manipulate `scope` variable within a JSP page.
- ❑ Tags in General-purpose library:



General-purpose tags 3-2

```
<%@ taglib  
    uri="http://java.sun.com/jsp/jstl/core"  
    prefix="c" %>
```

Directive inserted
to use General-
purpose tag

```
<body>
```

```
<c:set var= "example" value="{100+1}"  
    scope="session" />
```

Set tag is used
to set a variable
with a value and

Out tag displays the result of the value

```
<c:out value="{example}"/>
```

```
<c:remove var= "example" scope="session"/>
```

```
</body>
```

Remove tag is used to
remove the scoped variable

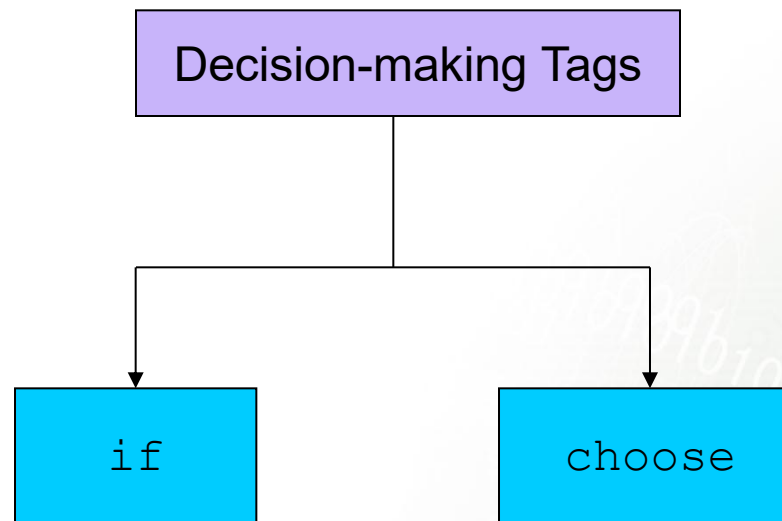
General-purpose tags 3-3

```
<body>
  <H1>Welcome to Shop Stop</H1>
  This JSP stores the sessionvariable in a session-
  scoped variable where the other JSPs in the web
  application can access it.
  <p>
    <c:set var = "sessionvariable" value = "${80+8}"
    scope = "session" />
    Value of session variable before removing it:
    <c:out value = "${sessionvariable}"/>
    <c:remove var = "sessionvariable" scope =
    "session" />
    <br/>
    Value of session variable after removing it:<br />
    <c:out value = "${sessionvariable}"></c:out>
  </body>
```

Demonstration: Example 1

Decision-making tags 2-1

- ❑ JSTL has decision-making tags to support conditions in a JSP page.
- ❑ Decision-making tags are:



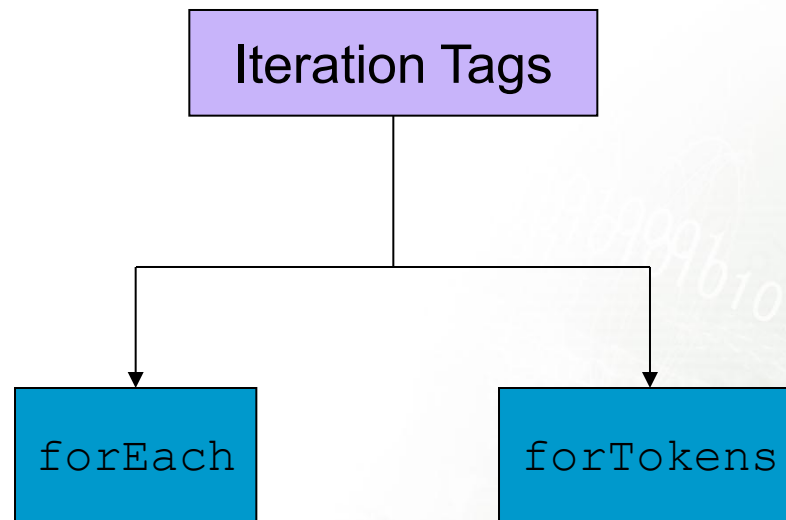
Decision-making tags 2-2

```
<body>  
  <c:if test = "${pageScope.signalStrength < 5}">  
    <c:set var = "signalFailure" value = "true"  
      scope = "page" />  
  </c:if>  
  <h1> Using If and Choose </h1>  
  <c:choose>  
    <c:when test = "${pageScope.signalFailure ==  
      true}">  
      Signal Failed  
    </c:when>  
    <c:otherwise>  
      Signal On  
    </c:otherwise>  
  </c:choose>  
</body>
```

Demonstration: Example 2

Iteration tags 3-1

- ❑ Iteration tags are used to evaluate the body of the tag multiple times.
- ❑ Tags in Iteration library are:



Iteration tags 3-2

```
<%@ taglib uri = "http://java.sun.com/jstl/core_rt" prefix  
    = "c" %>  
<c:set var = "language" value = "Java:J2EE:JSP:VB" scope =  
    "page" />
```

```
<body >
```

```
    <c:forEach var = "company" items = "<%= names %>"> ${company}  
    <br/>  
    </c:forEach>  
    <br/>
```

```
    <c:forTokens items = "${pageScope.language}" delin  
        var = "currentName" varStatus = "status">  
        Product No #P000<c:out value = "${status.count}" /> is  
        <c:out value = "${currentName}" /> <br />  
    </c:forTokens>
```

```
</body >
```

forEach tag iterates over a collection of objects. Numbers 1 to 4 will be displayed

forTokens iterates over the values separated by the supplied delimiters. It will display all the names

Iteration tags 3-3

❑ *Using Iteration tags in JSP page*

```
<body >
  <H1>Display Products & Company</H1>
  <c:forEach var="company" items="<%= names %>">
    ${company} <br/>
  </c:forEach>
  <br/>
  <c:forTokens items="${pageScope.language}"
  delims=":;|" var="currentName" varStatus="status">
    Product No #P000<c:out value="${status.count}" />
    is <c:out value="${currentName}" /> <br />
  </c:forTokens>
</body >
```

Demonstration: Example 3

I18N & Formatting Tag Library 4-1

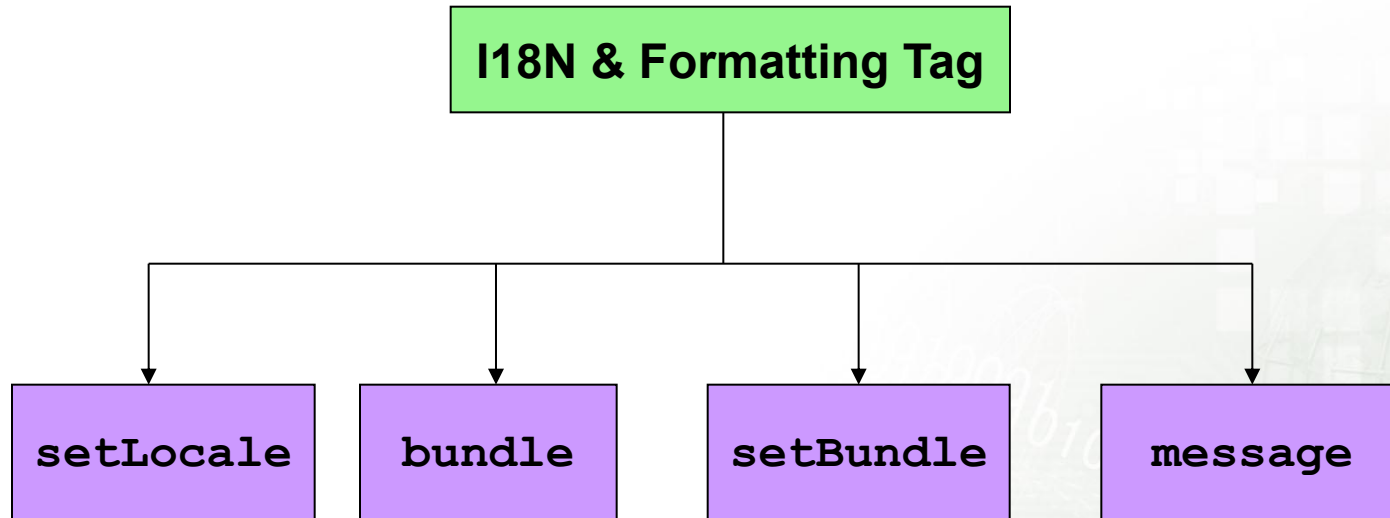
- ❑ The I18N and Formatting Tag Library helps in creating internationalized Web applications.
- ❑ It also helps standardize formatted numeric and date-time output.
- ❑ The features of an I18N application are:
 - ❑ The application can run worldwide
 - ❑ It supports multi-languages
 - ❑ It can be localized quickly

Syntax:

```
<%@ taglib uri =  
"http://java.sun.com/jstl/fmt_rt" prefix = "fmt" %>
```


I18N & Formatting Tag Library 4-2

- ❑ Tags in Internationalization (I18N) and Formatting Tag Library are:



I18N & Formatting Tag Library 4-3

- ❑ *Using I18N Tag Library in JSP page*
- ❑ *Usage of setLocale tag*

```
<body>
  <H2>Number Format Example </H2>
  Formatting <B>123.4</B> becomes :
  <fmt:formatNumber value="123.4" type="number"
  minFractionDigits="3" /><BR>
  <HR>
  <H2>Currency Format Example </H2>
  <c:set var="salary" value="125"/>
  Salary: <c:out value="\${salary}" /><BR>
  <fmt:setLocale value="en_GB"/>
  Formatting salary with Locale <B>en_GB</B> becomes :
  <fmt:formatNumber type="currency" value="\${salary}" /><BR>
  <fmt:setLocale value="en_IN"/>
  Formatting salary with Locale <B>en_IN</B> becomes :
  <fmt:formatNumber type="currency" value="\${salary}" /><BR>
  <HR>
</body>
```

Demonstration: Example 4

I18N & Formatting Tag Library 4-4

```
<c:if test="${param.language == 'en'}">
  <fmt:setLocale value="en" scope="session" />
</c:if>
<c:if test="${param.language == 'sv'}">
  <fmt:setLocale value="sv" scope="session" />
</c:if>
<c:if test="${param.language == 'de'}">
  <fmt:setLocale value="de" scope="session" />
</c:if>
<fmt:setBundle basename="labels" scope="session" />
```



Creates properties for the Resource Bundle

```
title=Industry Trends
```

```
title=Industri Trender
select_language=Välj språk
new_language=Nytt Språk
english=Engelska
swedish=Svenska
german=Tyska
submit=Rösta
```

```
<input checked="" type="radio" value="english" />
```

```
<fmt:message key="german" /><br>
```

```
<p>
```

```
<input type="submit" value="<fmt:message key='new_language' />" />
```

```
</form>
```

```
</body>
```

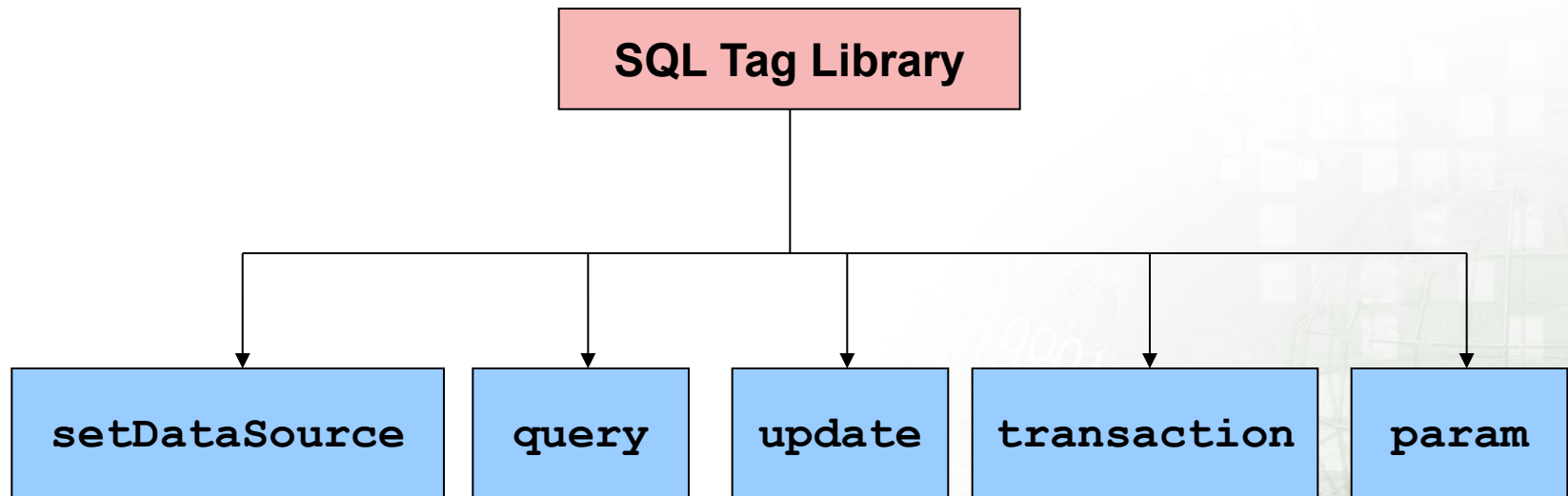
Demonstration: Example 5

SQL Tag Library 3-1

- ❑ SQL tag library is used to access and update database from a JSP page.
- ❑ The functions of an SQL Tag Library are:
 - ❑ Passing database queries
 - ❑ Accessing query results
 - ❑ Database modifications
 - ❑ Performing database transactions

SQL Tag Library 3-2

❑ Tags in SQL Tag Library are:



SQL Tag Library 3-3

```
<sql:transaction dataSource="${conn}">
    <sql:update var="newTable">

CREATE TABLE Products(
    ProductId int IDENTITY (1000, 1) NOT NULL
    primary key,
    ProductName varchar (20) NOT NULL ,
    ProductType varchar (15) NOT NULL ,
    Price varchar (5) NOT NULL ,
    Brand varchar (25) NOT NULL ,
    Description varchar (50) NOT NULL
)
    </sql:update>
</sql:transaction>
<sql:update var="newrow" dataSource="${conn}">
    INSERT INTO Products(ProductName, ProductType,
    Price, Brand, Description) VALUES('Jeans', 'Clothes',
    '1000', 'Lee', 'Good Quality Jeans')
    </sql:update>
    <sql:query var="products" dataSource="${conn}">
    select * from Products
    </sql:query>
```

Summary

- ❑ JSP Standard Tag Library (JSTL) tags are used to work with XML tags, access SQL database, internationalization tags.
- ❑ General-purpose tags are used to display, set, and remove the values found within a JSP page.
- ❑ Decision-making tags are used for conditional execution of the tags
- ❑ Iteration tags are used to evaluate the body of the tag multiple times
- ❑ I18N and formatting helps in creating internationalized Web application and standardize formatted numeric output.
- ❑ SQL tags are used to access and update database for low volume Web based application.